

Supplementary Material — Part E

Human-rater study

*What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports
Commentary*

The survey protocol and instrument as administered.

Sections in this part

S2 Human-Rater Survey Protocol and Instrument

S2 — Human-Rater Survey Protocol and Instrument

This supplement documents the design and administration of the human-rater validation study. It covers the intake question, task structure, sentence presentation, distractor design, domain definitions, and the reliability analysis framework. Actual survey responses are released separately as de-identified CSV files.

1. Overview

The human-rater study evaluates the 34-cluster taxonomy through two complementary evidence sources: **inter-rater reliability** (IRR) and **category recognition**. Nine raters with varying self-reported F1 familiarity (3 at level 2, 3 at level 3, 2 at level 4, 1 at level 5) completed a structured survey. The survey instrument was delivered via Google Forms and comprised one intake question followed by 34 cluster sections (one per cluster), each containing five tasks. Total mandatory questions: $1 + 34 \times 5 = 171$.

Parameter	Value
Raters (N)	9
Clusters evaluated	34
Tasks per cluster	5
Estimated time per rater	60–75 minutes
Platform	Google Forms
Total mandatory questions	171

2. Intake Question

Before the cluster sections, each rater answered:

“How familiar are you with Formula 1 broadcasts?”

Responses used a five-point ordinal scale:

Level	Wording
1	No prior experience with F1 broadcasts.
2	Occasional exposure; familiar with basic concepts (overtakes, pit stops).
3	Occasional viewer; recognise some drivers and teams.
4	Regular viewer; follow the current season actively.
5	Long-term follower; familiar with technical and strategic terminology.

3. Sentence Presentation

Each cluster section presented five sentences: four sampled from the target cluster and one out-of-cluster intruder. Sentences were drawn from the highlights corpus and shown in **original wording** (not entity-masked), preceded by a disclosure note that the underlying clustering analysis was performed on entity-masked text. Sentence order was randomised once at generation time (RNG seed = 42); the intruder position varied across sections (distribution across positions *a–e*: 6, 6, 8, 6, 8).

4. Task Order (per cluster section)

Tasks were presented in the following fixed order to enforce a **blind free-label** (Task 2 precedes Task 3 so that candidate labels do not prime the rater’s own description):

#	Task	Description	Chance
1	Intrusion detection	Pick the sentence that does not belong (5 choices + “All seem to belong”)	$\frac{1}{6} \approx 16.7\%$
2	Free label	Describe the theme in 1–4 words (free text; collected before label choices are shown)	—
3	Label selection	Choose the best label from 3 options + “None of these fit”	$\frac{1}{4} = 25.0\%$
4	Domain assignment	Choose the best broad domain from 4 options + “Doesn’t fit any”	$\frac{1}{5} = 20.0\%$
5	Importance rating	Rate the topic’s importance for understanding a race on a 1–5 Likert scale	—

Blindness enforcement. Each cluster section was split across two pages with a mandatory page break between Task 2 and Task 3. Page 1 showed the five sentences, Task 1, and Task 2; Page 2 re-displayed the same five sentences and then Tasks 3–5. This prevented candidate labels from being visible while the rater was filling in their free-label response.

5. Domain Options (Task 4)

Domain	Description shown to raters
Race dynamics	What happens on track (overtakes, defending, corners, starts)
Narrative & meta	Storylines, outcomes, records, championships, podiums
Strategy & technical	Race conditions and decisions (tyres, pit, pace, gaps, weather, damage)
Regulatory	Race control and rules (penalties, flags, safety car)
Doesn’t fit any	—

6. Importance Scale (Task 5)

1–5 Likert with anchors: 1 = *minor background element*; 5 = *central to understanding the race narrative*. The scale is ordinal; mean values per cluster are reported descriptively.

7. Distractor Design (Task 3 — label selection)

For each cluster, the three label options were: (1) the researcher-assigned thematic label; (2) the label of the nearest same-domain cluster by centroid cosine; (3) the label of the second-nearest same-domain cluster. Where the same-domain pool was too small or thematically too narrow (≈ 6 of 34 sections), distractors fell back to the nearest other-domain clusters. For C2 and C7 (both start-related), the near-synonym distractor was swapped to a non-start same-domain cluster to avoid near-identical choices. The position of the researcher label ($a/b/c$) was shuffled across sections; the ground-truth `ours_letter` field records the actual position used.

8. Intruder Selection (Task 1)

The intruder sentence was drawn from a different broad domain where possible (cross-domain intruder). In 5 of 34 sections where cross-domain candidates were exhausted, the intruder was drawn from a same-domain cluster instead. The ground-truth `intruder_letter` records the shuffled position of the intruder sentence.

9. Reliability Analysis Plan

IRR was computed per task across all nine raters:

- Krippendorff’s α with bootstrap 95% confidence intervals (Hayes & Krippendorff, 2007)
- Gwet’s AC1 as a prevalence-corrected complement (Gwet, 2008)

- Interpretive thresholds: $\alpha \geq 0.667$ tentative agreement, $\alpha \geq 0.800$ high (Krippendorff, 2004)

Tasks 1 and 4 reached the tentative-agreement threshold ($\alpha = 0.671$ and $\alpha = 0.687$ respectively); Task 3 ($\alpha = 0.516$) is treated as descriptive evidence only.

10. Data Availability

The full survey instrument (170 sentences across 34 sections) is not redistributed because of broadcast copyright. One illustrative section is included in the released code repository. De-identified response data (letter choices and importance ratings, no free text) and the analysis scripts are released openly. The full instrument is available upon reasonable request for non-commercial research use.